Table 1: Prespecified hyperparameter-budget outcomes. Cells report $10^3\times$ RMSE (mean ± seed SD), then the varied hyperparameter and measured minutes. DPI/PINN–HTE are frozen historical 960-state whole-function results; Deep BSDE is evaluated on the frozen 64-state subset and its time is cumulative training of 64 fresh scalar models. The protocols are therefore not identical paired evaluations.

Method	Sweep	B6 Full-Rank HJB			Dense Fourier		
		Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
		137.44 ± 12.94	184.31 ± 67.89	149.80 ± 57.85	110.78 ± 33.30	63.19 ± 14.61	61.80 ± 14.73
DPI	C0	80; 6.2 min	320; 24.4 min	450; 32.4 min	80; 4.7 min	320; 18.5 min	450; 26.0 min
		161.59 ± 39.57	132.75 ± 25.86	124.16 ± 33.83	76.70 ± 3.76	115.15 ± 61.09	122.35 ± 40.05
DPI	C1	512; 5.1 min	32768; 10.9 min	131072; 22.5 min	512; 4.1 min	32768; 8.6 min	131072; 22.4 min
		142.70 ± 18.04	108.17 ± 20.90	112.96 ± 30.47	110.78 ± 33.30	63.95 ± 8.79	77.65 ± 24.62
DPI	C2	4; 3.3 min	64; 9.6 min	256; 28.8 min	16; 5.0 min	128; 16.2 min	256; 29.6 min
		217.92 ± 54.33	186.23 ± 52.37	92.14 ± 40.03	70.34 ± 9.48	87.66 ± 8.56	105.20 ± 48.38
PINN–HTE	C0	30000; 3.3 min	120000; 13.2 min	240000; 26.4 min	30000; 3.4 min	120000; 13.5 min	240000; 26.9 min
		155.85 ± 88.23	199.63 ± 18.14	184.97 ± 94.40	103.54 ± 38.04	87.51 ± 11.34	76.91 ± 5.44
PINN–HTE	C1	64; 7.1 min	256; 26.6 min	512; 27.4 min	4; 3.4 min	64; 7.3 min	128; 8.6 min
		217.92 ± 54.33	136.12 ± 28.41	102.69 ± 12.21	93.72 ± 20.09	77.88 ± 19.71	68.06 ± 10.30
PINN–HTE	C2	512; 3.4 min	4096; 25.6 min	16384; 100.7 min	256; 3.3 min	2048; 13.4 min	4096; 25.8 min
		164.78 ± 0.88	160.08 ± 1.01	144.70 ± 0.61	76.67 ± 3.28	74.11 ± 0.81	59.49 ± 1.39
Deep BSDE	C0	1000; 21.5 min	2000; 43.0 min	4000; 85.8 min	1000; 22.6 min	2000; 45.1 min	4000; 89.9 min
		161.23 ± 1.51	160.08 ± 1.01	158.10 ± 0.26	74.60 ± 1.91	74.11 ± 0.81	66.94 ± 0.76
Deep BSDE	C1	32; 42.2 min	64; 43.0 min	128; 45.9 min	32; 44.5 min	64; 45.1 min	128; 44.3 min
		147.12 ± 0.69	160.08 ± 1.01	167.06 ± 0.83	65.03 ± 1.58	74.11 ± 0.81	76.71 ± 2.87
Deep BSDE	C2	10; 22.1 min	20; 43.0 min	40; 81.9 min	10; 23.2 min	20; 45.1 min	40; 88.2 min